

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-ac626a81867ecd9a8.jsonl`  
- SHA-256: `8b97f05274d2c39d8d93676c6afb5bd36a80675bafed49940f76e1f64bf66670`  
- Source modified: `2026-07-02T02:59:32+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:58:05.143Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer:

```
{  
  "title": "DOC_STATUS register + docs/README index are ~15 PRs behind the cleanup-audit and  
decision-snapshot trackers",  
  "category": "doc-rot",  
  "severity": "P2",  
  "file": "docs/DOC_STATUS.md",  
  "line": 47,  
  "evidence": "DOC_STATUS.md:47 says CODE_CLEANUP_AUDIT is 'Phase 1: P0/P1/P4/P6a shipped...  
Quick wins (Q4/Q5/Q6/Q8) next', but the audit doc itself (CODE_CLEANUP_AUDIT_2026-06-24.md:8,  
:180-205) shows Phase 1 COMPLETE plus Phase-2 items P8/P9/P11-P15/P17 all merged (#373-#386).  
DOC_STATUS has no row at all for DECISION_SNAPSHOT_REGRESSION.md (created 06-25, merged as #383)  
despite §0 saying to update on every lifecycle change; DOC_STATUS was last touched 2026-06-30  
(#403) without fixing this. docs/README.md:41 still calls the audit 'DRAFT, owner-gated' and  
README.md:42 says the snapshot doc's 'decisions in §2 need locking' though D1-D11 were LOCKED  
2026-06-25 (DECISION_SNAPSHOT_REGRESSION.md:26). NEXT_STEPS.md:6 still says 'owner M0 design  
sign-off pending', contradicted by its own line 14 ('M0-M9 LARGELY SHIPPED... ADR-014  
ratified').",  
  "why_it_matters": "DOC_STATUS is by convention the archival/lifecycle board and README is the  
ramp-up index; both are read at every session start. When they contradict the per-doc source-of-  
truth trackers, sessions either relitigate finished work or trust stale gating claims. The  
cleanup audit's own DoD (line 225) is blocked on exactly this README update.",  
  "proposed_fix": "One docs-only reconcile PR: refresh DOC_STATUS.md:47 row (Phase 2 nearly  
done, remaining = P10 + P16+), add a DECISION_SNAPSHOT_REGRESSION row, fix docs/README.md:41-42  
blurbs, and rewrite the NEXT_STEPS.md:6 'Where we are' paragraph to the post-06-21 state.",  
  "effort": "S",  
  "tracked_in": "docs/CODE_CLEANUP_AUDIT_2026-06-24.md §8 DoD (unchecked 'docs/README.md  
updated' item)"  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer,
C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-
data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator,
C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field;
non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup',
C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited
code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent
commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete
failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it

owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training, product/strategy/licensing, real cloud spend, counsel ToS/DPA?
HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you personally saw the defect/debt in current code; if you cannot reproduce the claim from the code, say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:58:12.463Z)

```
1      # Documentation Status & Archival Register
2
3      > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4      > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5      > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7      ---
8
9      ## 0. How to use this
10     - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11     - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12     - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13
14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git
mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.
23     5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).
24     6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/md5), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the
completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED**
(#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local
```

? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |

34 | `RALLY\_QUALITY\_RESEARCH.md` | Foundation, ****tracked & ACTIVE**** (\$7) | No | Drives the R-series; complete when \$7.1 single quality number is attributed |

35 | `R2\_EVAL\_METRIC\_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |

36 | `R1\_MILESTONE\_LAUNCH\_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |

37 | `RALLY\_DETECTION\_GAP\_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |

38 | `RALLY\_DETECTION\_FIXES\_PLAN.md` | ****DONE**** ? findings A/B/C resolved; P2b promoted `inactivity\_temporal\_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + \$7.1 number) | Archive with the R-series when the track wraps |

39 | `RALLY\_DETECTION\_CODE\_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |

40 | `PER\_VIDEO\_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |

41 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

42 | `EXPERIMENT\_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |

43 | `PROMINENT\_COURT\_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |

44 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |

45 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |

46 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora **vendor** docs archived ? `archives/roora-vendor/`. |

47 | `CODE\_CLEANUP\_AUDIT\_2026-06-24.md` | ****IN PROGRESS**** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next. | No | Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify |

48 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

49 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

50 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

51 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

52 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |

53 | `POST\_166...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

54

55 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")**

56 | Doc | Status | Note |

57 | ---|---|---|

58 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |

59 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

60 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

61 | `HOSTING\_PLAN.md` | PROPOSAL | ****?** partly overtaken by ADR-013 + the shipped site ? verify (finding #5)****** |

62 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |

63 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

64 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

65 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` /

`KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

66

67 ### 3c. Reports & terminal records (archive candidates once their track wraps)

68 | Doc | Status | Note |

69 |---|---|---|

70 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

71 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

72 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |

73 | `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` · `HARDENING\_LOOP\_HANDOFF.md` · `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? ****top-level pointer stubs**** (archived copies exist) | ****? confirm intentional stubs (per #165) vs finish-archiving (finding #4)**** |

74

75 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

76 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · ****`CREATING\_REELS.md`**** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · ****`CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md`**** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

77

78 ### 3e. Already archived (`docs/archives/`)

79 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past\_projects/low-regret-moves/` · ****`field-pmf/`**** (field-ops + customer-feedback, 2026-06-21) · ****`roora-vendor/`**** (annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md` · `HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

80

81 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

82 ****ADR-001 ? ADR-014**** (14, all ratified). The compute/serving lineage: ****ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014****. ADR-014 was ****RATIFIED 2026-06-21**** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

83

84 ## 4. Open due-diligence findings / actions (2026-06-21 review)

85

86 > ****? Doc-restructuring done (2026-06-21, owner-directed):**** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML ****data layer**** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

87 1. ****? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes `PER\_VIDEO\_OUTPUT\_LAYOUT.md`.**** Folded forward the `video\_slug` naming option (workspace §8.5) + the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

88 2. ****? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed**** with a current 2026-06-21 state + a ****cross-track prioritized pending list****; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

89 3. ****R-series / quality cluster is ACTIVE**** (`R1`,`R2`,`RALLY\_DETECTION\_GAP\_CLOSURE`,`RALLY\_QUALITY\_RESEARCH`,`QUALITY\_ITERATIONS`,`REAL\_FOOTAGE\_VALIDATION\_REPORT`,`RALLY\_DETECTION\_QUALITY\_REPORT`). ****Do NOT archive yet**** ? archive as a ***cluster*** once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

90 4. ****Confirm the COMPLETED top-level stubs**** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`,`HARDENING\_LOOP\_HANDOFF`,`DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.

91 5. ****`HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken**** by ADR-013 (DO dropped) + the shipped khelsutra site. ****? `UI\_PROPOSAL.md` banner added (2026-06-21)**** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md` localization track is likewise in implementation there. ****`HOSTING\_PLAN.md` banner still pending**** (ADR-013 already noted in it).

92 6. ****? RESOLVED (2026-06-21) ? `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED**** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized

(owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

93
94 ## 5. Maintenance
95 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
96

3. user (2026-07-02T02:58:13.661Z)

1 - RALLY_QUALITY_RESEARCH.md ? **Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a **shadow** signal), an ***adversarially-verified, cited external-research synthesis*** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a ***prioritized cheap?durable roadmap with falsifiable success criteria***, and (§7) a ***tracked project plan*** ? work-item checklist, sequencing, the single quality ***number*** attributed to the effort, and a ***definition of done*** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.

2 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap ***per video, before launch***, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is ***detection***: (G1) recall (local & Gemini both miss ~half on hard clips, and **different** rallies ? complementary), (G2) precision (local over-fires), (G3) ***rally-END over-extension on the quick-return-after-point*** (shuttle still moving ? needs a **player-state** signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A wiring, complementary fusion. ***Peer*** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

3 - PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the ***multicourt G2 over-fire***: identify the ***prominent (foreground) court*** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` ***before*** windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? ***pseudo-3D play-volume*** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

4 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

5 - EVALUATION.md ? How we measure and improve quality.

6 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with ***zero production regression*** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

7 - QUALITY_ITERATIONS.md ? ***Living log*** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a ***signal-fusion framework*** for rally detection ? producers emit confidence-scored ***signals*** at three temporal ***scopes*** (frame detectors / window analyzers / session analyzers), a learned ***scorer*** weights them, and an auditable ***report-first reconciliation*** pass lints the decision against the evidence. The spine: ***no special-casing in the decision**

path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**;
Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline** a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10

11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started)**: video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **\$7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

12

13 ## ? Documentation status & governance

14 - DOC_STATUS.md ? *** living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

15

16 ## Product / platform plans

17 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? *** Active tracked project (2026-06-21, `DRAFT` owner-gated)**: the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). \$7 M0?M10 tracker is the source of truth.

18 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ?
DELIVERED + ARCHIVED (2026-06-21): Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).

19 - PACKAGING_PLAN.md ? *** Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative**. \$7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). *** Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

20 - DESKTOP_APP_PLAN.md ? *** Superseded narrative (by `PACKAGING\_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app)**: Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-

feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

21

22 ## Practical / ongoing

23 - CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a §6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [README.md](../README.md); packaging roadmap ? PACKAGING_PLAN.md.

24 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, \$0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify\_compute\_path.py`) ? \$0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? ../deploy/cloudrun/README.md.

25 - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [deploy/gcp/README.md](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/\*` scripts are provider-agnostic and reused on GCP.

26 - DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khealsutra-rally-corpus` (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:/F:/Drive` ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

27 - [deploy/gcp/nightly_regression/README.md](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: NIGHTLY_ENGINE_REGRESSION.md). The lighter CPU cached-trajectory tripwire is `scripts/nightly\_regression.py` (local Windows Scheduled Task, PR #202).

28 - TRACKNET_WSL_SETUP.md

29 - Evaluation harness and calibration drivers (in `backend/eval/`)

30

31 ## Data layer (collection · annotation · golden sets ? **no ML**)

32 - [data_pipeline/](data_pipeline/README.md) ? **the isolated data pillar:** how footage is recorded (`VIDEO\_RECORDING\_GUIDELINES`), golden sets are created + which videos (`GOLDEN\_SET\_IMPLEMENTATION` · `GOLDEN\_SET\_PHASE1\_VIDEOS` · `GOLDEN\_VIDEOS`), and data is shared/ingested (`GOLDEN\_DATA\_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE\_DECOUPLED\_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.

33

34 ## Conventions & templates

35 - PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (??/??) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY\_QUALITY\_RESEARCH.md` §7 and `GOLDEN\_REGRESSION\_FIXTURES.md`. Use for any substantial design/project work.

36

37 ## Historical documents

38 Completed plans, old research, ADRs, and checkpoints live in archives/.

39

40 - CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`

for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)

```
41 - [CODE_CLEANUP_AUDIT_2026-06-24.md](CODE_CLEANUP_AUDIT_2026-06-24.md) ? **? Tracked
project (DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs,
tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config
migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. $7 progress tracker is
the source of truth; sibling of the completed hardening sweep above.
42 - [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? **? Tracked
project (DRAFT, owner-gated ? decisions in $2 need locking):** a deterministic **quality-neutral
proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat
plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-
AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel**
(non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`)
from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`.
$7 tracker is the source of truth.
43
44 - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ?
[archives/field-pmf/](archives/field-pmf/) (archived 2026-06-21; PMF validation still an
active *business* thread ? see `NEXT_STEPS.md`).
45 - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ?
[archives/roora-vendor/](archives/roora-vendor/) (archived 2026-06-21).
46 - Superseded: per-video output layout ? [archives/PER_VIDEO_OUTPUT_LAYOUT-
SUPERSEDED-2026-06-21.md](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (?
`PER_VIDEO_WORKSPACE.md`).
47
48 See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.
```

4. user (2026-07-02T02:58:23.592Z)

```
1 <!--
2 PROJECT DOC ? tracked code-cleanup + audit remediation plan.
3 Follows the PROJECT_DOC_TEMPLATE.md conventions.
4 -->
5
6 # Code Cleanup ? 2026-06-24 Deep Audit Remediation
7
8 > **Status:** `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes ·
**Owner:** Avi · **Created:** 2026-06-24 · **Last updated:** 2026-06-25 (DoD/line-count accuracy
fixes ? see Changelog)
9 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
10 > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
11 > **Relation to existing docs:** Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (the
2026-06-10 hardening sweep).
12 >
13 > **Phase 1 summary:** All structural refactorings landed. `backend.api.job_worker`
lifted from mypy quarantine. Suite: **1463/1463 green**. `ruff`: all checks passed. `mypy`:
quarantined modules unchanged (ratchet held).
14
15 ---
16
17 ## 0. TL;DR
18
19 A deep, whole-repo audit on 2026-06-24 surfaced **35 items** (33 from fresh review + 2
absorbed from `docs/BACKLOG.md`) across bugs, technical debt, operational risks, and quick wins.
This doc organizes them into a phased cleanup project: **Phase 1 = pure refactorings** (no logic
change, all tests must stay green, coverage must not drop), **Phase 2+ = logic fixes +
hardening**. The first approved targets are **B2, D3, D2, B5** ? the `Success`/`Failure`
contract for Gemini, typed-config migration completion, app-factory refactor, and the no-op
segmenter validator fix.
20
21 ---
22
```



```

23     ## 1. Problem & goal
24
25     The codebase has grown rapidly since the June 2026 scaffolding and has accumulated
several patterns that make it harder to extend safely:
26
27     - Silent failure masking ? `analyze_video()` returns `[]` on ANY error (missing API
key, network failure, model crash) indistinguishable from "video has no rallies."
28     - Dual config access ? new code uses typed `AppConfig` attribute access; old code
(validators, segmenters) uses legacy `.get("indexing", {}).get(...)` dict chains. The
`extra='allow'` silently keeps typo'd config keys.
29     - Optional module globals ? `db_instance` and `global_config` are `Optional[...] =
None`, set only in `main()`. Importing `app` without `main()` ? NPE on every endpoint. The mypy
quarantine on `backend.main` and `backend.api.job_worker` exists because of this.
30     - No-op validator ? `segmenter_must_be_registered` field validator is a pass-
through; a typo'd segmenter name in `config.json` passes validation and fails at runtime.

```

5. user (2026-07-02T02:58:24.486Z)

```

170     - A green suite after Phase 1 does NOT mean the code is "clean." It means the
structural debt targeted by D2/D3/B5 is resolved. The remaining 28 items still need attention.
171
172     ---
173
174     ## 6. Deliverables & progress tracker ? source of truth
175
176     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row,
branched from `origin/master`, no stacking.
177
178     | ID | Deliverable | Depends on | Gated? | Status | PR |
179     |---|-----|-----|-----|-----|---|
180     | P0 | Docs-only PR ? this project doc** + index in `docs/README.md` | ? | Owner
approval | ? | [352](https://github.com/Khelsutra/badminton-highlight-indexer/pull/352) |
181     | ? | WASB cache test fix (Windows) ? 2 pre-existing failures | ? | No | ? |
[353](https://github.com/Khelsutra/badminton-highlight-indexer/pull/353) |
182     | P4 | B5: Fix `segmenter_must_be_registered` validator | P0 | No | ? |
[354](https://github.com/Khelsutra/badminton-highlight-indexer/pull/354) |
183     | P1 | D3-a: Convert validators to typed `AppConfig` (6 files) | P0 | No | ? |
[355](https://github.com/Khelsutra/badminton-highlight-indexer/pull/355) |
184     | P6a | Q3: Deduplicate `_default_db_path` | P0 | No | ? |
[356](https://github.com/Khelsutra/badminton-highlight-indexer/pull/356) |
185     | P6b | Q4: Cache builtin defaults | P0 | No | ? |
[358](https://github.com/Khelsutra/badminton-highlight-indexer/pull/358) |
186     | P6c | Q5: requirements.txt torch comment + Q6: isort (I) + Q8: cv2 comment |
P0 | No | ? | [359](https://github.com/Khelsutra/badminton-highlight-indexer/pull/359)
[360](https://github.com/Khelsutra/badminton-highlight-indexer/pull/360)
[361](https://github.com/Khelsutra/badminton-highlight-indexer/pull/361) |
187     | T1 | Tier 1: Q2 (motion return type) + D11 (codec docstring) | P0 | No | ? |
[362](https://github.com/Khelsutra/badminton-highlight-indexer/pull/362) |
188     | T2 | Tier 2: B6 (resolution_fps fallback + warning log) | P0 | No | ? |
[363](https://github.com/Khelsutra/badminton-highlight-indexer/pull/363) |
189     | T3 | Tier 3+4: D12 (sys.path hack) + B3 (PTS gap config) + O5 (lock note) + D7
(centralize TUNED) | P0 | No | ? | [364](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/364) |
190     | P6d | ~D7: Centralize hardcoded tuning constants~ ? folded into T3 | P0 | No | ?
| [364](https://github.com/Khelsutra/badminton-highlight-indexer/pull/364) |
191     | P2 | D3-b: Convert segmenters to typed `AppConfig` (15 files: 5 segmenters + 4
detectors + worker + eval + tools) | P1 | No | ? |
[366](https://github.com/Khelsutra/badminton-highlight-indexer/pull/366) |
192     | P3 | D3-c: Remove `.get()` compat shim; flip `extra='forbid'` (18 files,
+87/-169) | P2 | No | ? | [367](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/367) |
193     | P5 | D2: App-factory refactor ? replace module-level Optional globals with
`create_app(config, db)` ? FastAPI | P3, P4 | No | ? |
[370](https://github.com/Khelsutra/badminton-highlight-indexer/pull/370) |
194     | P7 | Phase-1 verification: full suite green, coverage ? 84%, mypy ratchet |

```

P1?P6 | No | ? | 1463/1463 pass (#371); `backend.api.job\_worker` lifted from mypy quarantine | 195 | ? | ? | ? | ? | ? | ? | ? |

196 | **P8** | **B2: Error-flagging in Gemini provider metrics** (Phase 2) | P7 | No | ? | |
 [#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |

197 | **P9** | **B1: Remove fabricated data from motion segmenter** ? random
 server/receiver/shot_count/speed ? `None`; events dropped | P7 | No | ? | |
 [#385](https://github.com/Khelsutra/badminton-highlight-indexer/pull/385) |

198 | **P10** | **D1: Split `main.py` god-module** (~1450 lines ? smaller modules) | P7 | No
 | ? | ? |

199 | **P11** | **O1: Gemini remote-file GC** ? per-call `\_safe\_delete` (#373) + startup
 orphan sweep scoped to our `UPLOAD\_TAG`-tagged files, grace-gated, HTTP-timeout-bounded | P7 |
 No | ? | | [#380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380) |

200 | **P12** | **D13: ByteTrack deprecation** ? migrated `sv.ByteTrack` ?
 `trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? | |
 [#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |

201 | **P13** | **D4: Deep config merge** ? section-level replace ? recursive merge | P7 |
 No | ? | | [#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |

202 | **P14** | **O2: Single-executor risk** ? documented with mitigations in job_worker.py
 | P7 | No | ? | | [#374](https://github.com/Khelsutra/badminton-highlight-indexer/pull/374) |

203 | **P15** | **Q7: Type-clean `motion.py`** ? ratchet exercise | P7 | No | ? | |
 [#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |

204 | **P17** | **B2 (cont.): finish the `Success`/`Failure` return contract** ? the 4
 callers that dropped provider error metrics (`trajectory\_hybrid`, `yolo\_hybrid`,
 `gemini\_refine.refine\_window`, `gemini\_refine.full\_video\_rallies`) now surface
 `metrics["error"]` loudly instead of silently treating a failure as "0 rallies". (Full
 `Success`/`Failure` return-type migration of the hybrids is follow-up.) | P8 | No | ? | |
 [#378](https://github.com/Khelsutra/badminton-highlight-indexer/pull/378) |

205 | **P16** | Phase 3 items (B4/B7/B8/D5/D6/D8-D10/D14/O3/O4) | P15 | No | ? | ? | |

206

207 ---

208

209 ## 7. Open questions

210

211 1. **Q: Should D1 (split `main.py`) be Phase 1 or Phase 2?` ? Leaning Phase 2: D2 (app-
 factory) is a natural precursor, and D1 is lower risk after the globals are gone. Owner to
 confirm.

212 2. ~~Q: Should Phase 1 include enabling `I` (isort) in ruff.toml?~~ ? ? Shipped in
 #360.

213

214 ---

215

216 ## 8. Definition of done

217

218 - [x] P0: This doc approved and merged to `master`

219 - [x] P1?P3: Typed-config migration complete ? no `.get()` dict access remains in
 backend code; `extra='forbid'` on `AppConfig`

220 - [x] P4: `segmenter\_must\_be\_registered` actually validates against the registry

221 - [x] P5: `create\_app()` injects shared state via `app.state` (handlers read
 `request.app.state`); `backend.api.job\_worker` removed from mypy quarantine. (The Optional
 module globals `db\_instance`/`global\_config` are **retained as a transitional back-compat shim**
 and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)

222 - [x] P6: Quick wins Q1?Q6, Q8 landed

223 - [x] P7: Full test suite ? 1463/1463 pass (#371); `ruff` + `mypy` clean;
 `backend.api.job\_worker` lifted from quarantine

224 - [x] All PRs are single-purpose, branched from `master`, with descriptive commit
 messages

225 - [] `docs/README.md` updated to reflect the project's terminal state

226 - [] Final status flipped to `DONE` and this doc moved to
 `docs/archives/past\_projects/`

227

228 ---

229

230 ## 8.1 ? Next items ? Phase 2 pick-up (prioritized)

231

232 All items below are **Claude-doable**, small-PR-friendly, and unblocked. Pick up in

```

order:
233
234     ### Immediate (HIGH impact, low risk)
235
236     | # | ID | What | Why |
237     |---|----|-----|-----|
238     | ~1~ | ~~~B2 / P17~~~ | ~~~Finish~~ the `Success`/`Failure` return contract~~ ?
***? DONE (this PR).** The 3 callers that dropped provider error metrics now surface
`metrics["error"]` loudly. **Follow-up remains:** migrate the hybrid segmenters' `process_video`
return type from bare `List[Dict]` to the ABC's `Result[...]` (`Success`/`Failure`) ? currently
suppressed with `type: ignore[override]` in `trajectory_hybrid.py` (motion.py is already fully
migrated). |
239     | ~2~ | ~~~O1 / P11~~~ | ~Gemini remote-file GC~~ ? ***? DONE (#380).** Per-call
`_safe_delete` (#373) + startup orphan sweep scoped to `UPLOAD_TAG`-tagged files, grace-gated,
HTTP-timeout-bounded. |

```

6. user (2026-07-02T02:58:33.048Z)

```

1      <!--
2      PROJECT DOC ? tracked design plan for a deterministic "quality-neutral proof".
3      Follows docs/PROJECT_DOC_TEMPLATE.md. $7 progress tracker is the source of truth.
4      -->
5
6      # Decision-Snapshot Regression ? a deterministic quality-neutral proof
7
8      > **Status:** `DRAFT` ? **$2 decisions LOCKED (2026-06-25); ready to start $7 P1** .
**Owner:** Avi . **Created:** 2026-06-25 . **Last updated:** 2026-06-25
9      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
10     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` once work starts.
11     > **Relation to existing docs:** **extends** `GOLDEN_REGRESSION_FIXTURES.md` (post-
detector fixtures) and the existing characterization snapshot in
`tests/test_golden_fixtures.py`; **peer of** the M4 quality-floor track and the GPU
`scripts/nightly_reelcount.py` e2e nightly.
12     > **Honesty note:** claims tagged `[verified]` (checked vs code) / `[design]`
(proposed).
13
14     ---
15
16     ## 0. TL;DR
17     A deterministic regression that proves a change is **quality-neutral** (behavior-
preserving) by freezing and comparing the pipeline's **decision output** ? the rally segment
list **+ the stitch/concat plan** ? run off a **cached trajectory** (GPU-free). It **extends**
the existing characterization snapshot (`test_replay_matches_committed_expected`, today
segmentation-only) down to the stitch plan, runs offline in CI, and gives the **behavioral
proof** that lets a *genuine* refactor (not just `ruff format`) be confidently called quality-
neutral. **Most important caveat: never hash the encoded reel `.mp4`** ? ffmpeg/x264/NVENC/mux-
timestamps make those bytes non-deterministic, so a reel-MD5 would false-alarm on every ffmpeg
bump.
18
19     ## 1. Problem & goal
20     "Presumably quality-neutral" changes (refactors) need a cheap, deterministic proof that
they actually preserve behavior ? so they can be merged (and auto-signed-off) with confidence.
Today we have two weaker tools: **AST-identity** [verified] (used to bless `ruff format` in
#379) only proves *formatting* is unchanged ? it cannot bless a real refactor (function
extraction, rename, loop reflow) that changes the AST but should preserve behavior; and the full
**real-video F1 / reel-count e2e** needs GPU + the gitignored corpus
(`scripts/nightly_reelcount.py`) [verified] so it can't gate a PR. The decision-snapshot fills
the gap: a deterministic, offline, behavior-level proof.
21
22     **What "good" looks like:** a refactor PR runs the snapshot in CI; byte-identical
decision output ? provably quality-neutral ? eligible for auto-sign-off; a behavior-changing
edit fails with a **readable diff**.
23

```

```

24      **Read first:** `tests/test_golden_fixtures.py::test_replay_matches_committed_expected`
[verified] (the existing snapshot ? its own message: "behaviour changed vs frozen snapshot
(characterization, not correctness)" ), `backend/eval/golden_fixtures.py`,
`backend/pipeline/stitcher/compiler.py` (the concat plan), audit doc **B1** ( `motion`'s random
fields), `scripts/nightly_reelcount.py` (the distinct GPU e2e).
25
26      ## 2. Decisions  *(D1?D11 **LOCKED** 2026-06-25)*
27
28      | # | Decision (recommendation) | Rationale | Status |
29      |---|-----|-----|-----|
30      | **D1** | **Snapshot the DECISION output, never the encoded reel.** Freeze the rally
segment list + the stitch/concat plan; never MD5 the reel `mp4`. | Encoded mp4 bytes are non-
deterministic (ffmpeg version string, x264 threading, NVENC, mux timestamps) ? a reel-MD5 false-
alarms on every ffmpeg bump/machine. The repo's own `nightly_reelcount.py` already asserts
count+metrics, **not** bytes [verified]. | **LOCKED** (2026-06-25) |
31      | **D2** | **Input = a cached trajectory (detector output), not raw video.** Run from a
pinned trajectory CSV (the `fx_r0X` fixtures). | The WASB detector (video?trajectory) is the
GPU/WSL + CV-float-non-deterministic upstream; caching its output makes the proxy deterministic
+ CI-runnable [verified] ? the fixtures are post-detector]. | **LOCKED** (2026-06-25) |
32      | **D3** | **Coverage = segmentation decision + stitch/concat plan; STOP before the
encode.** Extend the existing segmentation snapshot down to the concat manifest (clips, cut
points, order, padding). | Segmentation is already snapshotted [verified]; the stitch plan is
the deterministic missing layer. The encode is non-deterministic (D1). | **LOCKED** (2026-06-25)
|
33      | **D4** | **Compare = canonical-JSON snapshot with a readable diff** (an MD5 of the
canonical JSON may be a fast-path summary, but the field-level diff is the artifact). | A
refactor that trips it needs a debuggable diff, not "hash changed" ? mirrors
`test_replay_matches_committed_expected`'s mismatch report [verified]. | **LOCKED** (2026-06-25)
|
34      | **D5** | **Snapshot only deterministic segmenters/fields; exclude the random
fabricated fields.** Use a deterministic segmenter (CONTROL/fusion on trajectory) and a
`{start,end,shuttle}`-style schema that excludes `motion`'s random
`server/receiver/shot_count/avg_speed`. | `motion` fabricates random fields (audit **B1**)
[verified] ? non-snapshot-able as-is. **B1's fix (drop the fabrication) is the enabler that
would make `motion` snapshot-able. | **LOCKED** (2026-06-25) |
35      | **D6** | **The reel mp4 gets a FUNCTIONAL smoke, not a byte hash** (exists, ~duration
within tol, N segments, sane streams). | Corollary of D1 ? verify the encode is *sane*, not
*identical*. | **LOCKED** (2026-06-25) |
36      | **D7** | **Runs in per-PR CI, offline, CONTROL-only ? a refactor gate, not a quality
floor.** | The point is catching a behavior change in a refactor *before merge*; it must be
fast, GPU-free, on the pinned CONTROL config. | **LOCKED** (2026-06-25) |
37      | **D8** | **Intentional changes re-freeze WITH a documented rationale (Launch
Report).** A `--update-snapshot` flag exists, but a mismatch on an *intended* change must be re-
frozen with the Launch-Report rationale recorded. | Without the discipline a snapshot becomes a
rubber stamp; ties to the launch-report convention. | **LOCKED** (2026-06-25) |
38      | **D9** | **Scope boundary: this is the quality-NEUTRAL proxy ? distinct from the M4
quality-FLOOR (per-slug F1), Tier-2 real-video e2e (GPU), and `nightly_reelcount.py`. Keep
them separate + complementary. | Conflating "behavior unchanged" with "quality is good enough"
muddies both. | **LOCKED** (2026-06-25) |
39      | **D10** | **Deterministic-segmenters-only, ENFORCED by a determinism guard.** Cover
only deterministic segmenters; to *ensure* no non-deterministic one slips in (? a flaky
snapshot), the harness runs each fixture **twice and asserts byte-identical decision output**
before freezing ? any segmenter whose two runs differ (hidden `random`, set-ordering, float/time
non-determinism) is **rejected from snapshot scope** and flagged. Mechanical, not a hand-
maintained allowlist. | Owner, 2026-06-25. `motion` is non-deterministic until **B1 / #385**
lands; then it auto-becomes eligible (no allowlist edit). | **LOCKED** (2026-06-25) |
40      | **D11** | **Auto-sign-off integration is STAGED: ADVISORY first ? GATING later.** Ship
the snapshot as an advisory/informational check; graduate it to a **gating** check (gates a non-
AST-identical refactor's auto-LGTM) only after it's proven stable on real refactors. | Owner,
2026-06-25. Resolves S8 Q1 ? the strategic payoff, de-risked by starting advisory. | **LOCKED**
(2026-06-25) |
41
42      ## 4. Design / architecture
43      **Flow (all offline, deterministic):**
44      `pinned trajectory CSV` ? [segmentation: CONTROL segmenter] ? segment list ?

```

[deterministic stitch-PLAN builder: clips + cut points + padding + order, NO ffmpeg] ? canonical JSON {schema_version, segments[], stitch_plan{}} ? MD5 / field-diff vs frozen golden`.
45 Mismatch ? behavior changed (decide if intended; re-freeze via D8).

7. user (2026-07-02T02:58:34.081Z)

```
1      # Next steps (live) ? refreshed 2026-06-21
2
3      > **How to read:** this top section is the **current cross-track state + prioritized
pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
**owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it
is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5      ## ? Where we are (2026-06-21)
6      **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only**
(ADR-013, DO dropped). **COMPUTE_DECOUPLED_SERVING** subproject spun up (#272, **ADR-014**) ?
one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video
workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273).
The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /
owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8      > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is
**not** cleared for an external/commercial launch; we stay in **development**. This is the
expected default for the dev phase. Two unblockers gate a future GO: ** (1) ** set + meet the
**ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 toF1 ablation,
item 8); ** (2) ** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
9
10     ## ? Prioritized pending ? all tracks
11     **P0 ? highest leverage / unblocks the most**
12     0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): ** (i) ** deploy the engine as a reachable
**service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), ** (ii) **
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), ** (iii) ** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, ** (iv) ** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]].
13     0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ** (1) ** P3 **GPU
validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]` ; ?
native-WASB 0-rally tuned-config blocker), ** (2) ** P2b-install (uv/script + truly-local default
config), ** (3) ** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
`[[next-session-packaging-app]]`.
14     1. ? **COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014
ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged
(default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8
gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel
ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12
real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a
**$100/?10k cap** (D17).
15     2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus
training feed on it.
16     3. **[owner + Claude . GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).
17
```

```

18      **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
19      4. **[Claude] `COMPUTE_DECOUPLED_SERVING` batch ? REMAINING: M11 + M12**
(M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-
eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked
`consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** =
`backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme
(regex already allows the hyphen; add to `_KNOWN` + `_backend_class` + the StorageConfig
Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://`
untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text =
owner/counsel.
20      5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video
routing of `detector/+`tmp/` *[the `output/chunks_` de-hardcode base = ? done via #282]* ? P4
? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-
ledger migration ? author the version bump once.
21      6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM
helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN
active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
22      7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
23      8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the
ablation) ? needs GPU-extracted golden features.
24
25      **P2 ? later / gated / deferred**
26      9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
27      9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in
`RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED**
windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so
`inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in
#394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the
rally tail open.
28      10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
29      11. **Multisport** (table-tennis first, then pickleball).
30      12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
31      13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
32      14. **Rename repo off "badminton"** (owner, BACKLOG).
33      15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector
**`md5`** export (collector repo).
34      16. **PMF field validation** (owner-side, business track).
35
36      **Doc-hygiene (tracked in `docs/DOC_STATUS.md` $4):** ? #1 per-video reconcile + ? #2
this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.
37
38      ---
39
40      ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

```

8. user (2026-07-02T02:58:51.835Z)

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6e973af docs: fix stale rally-cluster claims (no archival ? cluster still gated) (#403)
bab72c6 docs: update cleanup project tracker ? Phase 1 items shipped (#357)
8e3aff1 docs: 2026-06-24 code-cleanup audit ? 33-item project doc (Phase 1 approval gate) (#352)
2f4ecad docs(runbook): GCS-bucket videos ? highlights on the L4 Cloud Run + committed build
config (#346)
2c1028a docs(guide): add CREATING_REELS.md ? end-user install?reel walkthrough (#343)
fefa412 docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project
(#275)
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/
(#276)
---
1283e18 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-neutral proof
(#383)
8e3aff1 docs: 2026-06-24 code-cleanup audit ? 33-item project doc (Phase 1 approval gate) (#352)

```

2f4ecad docs(runbook): GCS-bucket videos ? highlights on the L4 Cloud Run + committed build config (#346)
2c1028a docs(guide): add CREATING_REELS.md ? end-user install?reel walkthrough (#343)
df3b225 docs(nightly): one-stop operations manual for the nightly engine-regression (#325)

4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)
6e973af docs: fix stale rally-cluster claims (no archival ? cluster still gated) (#403)
cc60ebe Fix/rally detection edge cases (#394)
a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)
2afbadf fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

master...origin/master

9. assistant (2026-07-02T02:59:16.548Z)

All claims verified. Calling StructuredOutput.

10. user (2026-07-02T02:59:32.686Z)

Structured output provided successfully